

Nusrat Jabin Prova

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EDUCATION

Metropolitan University, Sylhet, Bangladesh
Bachelor of Science - Computer Science

Graduated Dec 2024
GPA: 3.64

WORK EXPERIENCE

Tutor

Jan 2022 - Feb 2025

Private tutor, Sylhet, Bangladesh

- Delivered personalized tutoring in math, science, and English to high school students through one-on-one and small group sessions
- Designed interactive lesson plans to improve academic performance, build confidence, and address each student's unique learning needs
- Fostered a supportive learning environment through effective communication, mentoring, and problem-solving, helping students achieve their academic goals

Digital Budget Manager

Apr 2023 - Oct 2024

A. Ali Enterprise, Sylhet, Bangladesh

- Created and managed budgets to ensure effective resource allocation, monitored expenses, and addressed financial variances through practical solutions
- Prepared comprehensive financial documents, including budget proposals, statements, and variance reports, to support data-driven decision-making

ENGINEERING PROJECTS

PM 2.5 Concentration Analysis (Machine Learning, Python Application)

Jul 2024 - Dec 2024

- Developed a robust predictive model for PM2.5 air pollutant concentration by integrating feature engineering techniques with an ensemble stacking linear regressor and Artificial Neural Networks (ANN)
- Applied advanced data preprocessing techniques including outlier detection, missing value imputation, normalization, and correlation-based feature selection to optimize model input
- Utilized machine learning models (Random Forest, Gradient Boosting, Extra Trees) combined with ANN to accurately predict PM2.5 concentrations and identify key contributing factors

Book Recommendation System (AI Project, Python Application)

Mar 2024 - May 2024

- Created a book recommendation system using user rating data from Kaggle to provide personalized suggestions
- Developed and tested the application using Jupyter Notebook and PyCharm for an efficient workflow
- Connected the application to a local database via PyCharm and localhost for seamless data processing and management

Irish Flowers Classification Using KNN Algorithm (Machine Learning, Python Application)

Aug 2023 - Dec 2023

- Developed a machine learning algorithm to analyze data from the Iris flowers dataset, sourced from Kaggle
- Applied the K-Nearest Neighbors (KNN) algorithm to process and analyze the dataset effectively

Improve Sylhet (Blog Application, Website)

Aug 2023 - Dec 2023

- Designed and developed a blog application to connect citizens with government agencies for enhanced civic engagement
- Implemented features allowing users to create accounts, share image-based posts, and tag relevant government teams to report local issues such as poor road conditions
- Enabled real-time tracking of reported issues, public awareness, and improved quality of life in Sylhet through community-driven action

Bluetooth Controlled Security Door Lock System (Microprocessor, C Application)

Jan 2023 - Apr 2023

- Created a Bluetooth-enabled security door lock system that authenticates valid users and unlocks automatically
- Programmed the automated door system using C and Arduino to ensure seamless functionality
- Utilized Arduino Uno, Bluetooth AC-05, Relay Board, and Solenoid Lock to develop and implement functional hardware systems

Digital Diary (Blog Application, Website)

Jan 2023 - Apr 2023

- Built a blog application that allows users to create and maintain digital journals
- Developed the application using the Laravel framework for a robust and efficient backend
- Leveraged tools like PHP, Java, HTML, CSS, and MySQL to design and implement the platform

Election System (Election Report, C Application)

Aug 2022 - Dec 2022

- Created an application for Metropolitan University to monitor and manage university election results
- Designed and implemented the application using the C programming language for efficiency and reliability
- Included an admin interface to allow administrators to update and manage election data seamlessly

TECHNICAL SKILLS

Programming Languages: SQL • Python • Java • HTML • CSS • C/C++ • JavaScript • Git • Emu8086 • Shell Scripts

Applications: GitHub, Visual Studio, PyCharm, IntelliJ, Eclipse, Codeblocks, LaTeX, Google Colab

Databases & Platforms: MongoDB, MySQL, Amazon's AWS S3, Firebase

Operating Systems: Windows, Linux (Mint), Mac OS, Chrome OS